

Soham Das

Full-Stack & GenAI Developer | AI Engineer

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SUMMARY

GenAI & Full-Stack Developer with hands-on experience designing and deploying AI-powered applications using **Python**, **RAG pipelines**, **LangChain**, and **OpenAI APIs**. Shipped 3 production projects spanning document intelligence, AI-powered healthcare, and real-time systems. Passionate about researching emerging AI/ML trends, building intelligent agents, and collaborating cross-functionally to deliver impactful, production-ready solutions.

SKILLS

GenAI & LLM Engineering: Python, RAG, LangChain, LangGraph, Prompt Engineering, Fine-tuning, Agent SDK, MCP, Agentic Workflow, Vector Databases, Graph Databases

AI/ML Concepts: NLP, Entity Extraction, Embedding Models, Retrieval-Augmented Generation, AI Pipeline Design, Model Evaluation, Algorithm Testing

Frontend & Web Development: React.js, Next.js, TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3, Redux, Bootstrap

Backend & APIs: Node.js, Express.js, REST APIs, GraphQL, Drizzle ORM

Databases & Cloud: MongoDB, PostgreSQL, MySQL, Firebase, MongoDB Atlas, Qdrant Cloud

Tools & DevOps: Git, GitHub, Docker, Postman, VS Code, Vercel, Render

PROJECTS

NuroDesk – AI-Powered Document Intelligence | *LangChain, RAG, Qdrant, React, Node.js* | [Live Demo](#) | [GitHub](#) | [Video](#)

- Architected a production **RAG platform** supporting **5 document formats** (PDF/OCR, CSV, TXT, web, YouTube) with **3,000+ vector embeddings**, 15-chunk retrieval, and per-user document isolation — a full AI solution built from design to deployment.
- Integrated **OpenAI GPT-4**, **Whisper** transcription, and **ada-002 embeddings** (1536-dim) with **Qdrant** vector DB; **tested and optimized** AI query latency to under 2s with 3–5 source citations per response.
- Engineered scalable **Python-based** AI ingestion pipelines with **LangChain** (1,000-token chunks, 200 overlap), **Tesseract OCR**, and **yt-dlp** — validated and stress-tested across **500+ documents**.
- Deployed on **Vercel** and **Render** with **MongoDB Atlas** and **Qdrant Cloud**; actively monitoring AI model behaviour and pipeline performance in a live multi-user production environment.

FastCare – AI-Powered Healthcare Platform | *Python, NLP Pipeline, Next.js, Node.js* | [Live Demo](#) | [GitHub](#) | [Video](#)

- Developed an end-to-end **AI healthcare solution** integrating **OpenAI GPT-4o** into a **Python NLP pipeline** to extract structured clinical entities — medications, lab values, allergies — from unstructured patient documents.
- Built an **automated AI contradiction engine** detecting drug-allergy conflicts and abnormal lab values — an intelligent validation layer that tests AI accuracy and surfaces critical health flags to clinicians instantly.
- Engineered the full-stack system (**Next.js** frontend, **Node.js** API, **MongoDB** schema) with Cloudinary storage and **NextAuth** authentication; collaborated as lead developer in a **cross-functional team** to ship a production AI product.
- Researched and implemented a **chronological medical timeline** feature surfacing AI-generated clinical summaries per patient — demonstrating applied research turning emerging AI capabilities into real-world tools.

FluxUI – AI-Powered UI Generation Platform | *OpenAI, Next.js, NeonDB, Drizzle ORM* | [Live Demo](#) | [GitHub](#) | [Video](#)

- Built a production-grade **AI solution** that converts natural language prompts into **multi-screen responsive UI layouts** using **OpenAI GPT models** — a direct application of AI for design automation and code generation.
- Designed and tested **AI-driven workflows** for screen generation, UI editing, structured layout creation, and theme-aware responsive design — validating model outputs against edge cases and visual accuracy benchmarks.
- Implemented **authentication**, **premium feature gating**, and **Stripe subscription payments** via Clerk — researched and integrated third-party AI and payment APIs to deliver a monetized, full-stack AI product.

EDUCATION

ST. THOMAS' COLLEGE OF ENGINEERING & TECHNOLOGY

B.Tech in Computer Science and Engineering | CGPA: 8.20/10 (till 5th Semester)

Kolkata, India

Aug 2023 – Present